

1 Basic Information

Term	Spring 2026
Course Title	電腦硬體與程式語言在行為科學實驗與大數據分析之應用 The Applications of Computer Hardware and Programming Languages in Behavioral Experiments and Big-Data
Course Code	NS5116
Classroom	S5-607-1
Credit Hours	3.0
Class Schedule	Thursdays 9:00 AM – 12:00 PM (make-up classes on 2026/05/12 and 2026/05/26, Tuesdays)
Course Teacher	Erik Chang 張智宏
Teacher Email	audachang@gmail.com
Contact Number	03-4227151#65209
Office and Location	Building: Science 5, Teacher Room: 601
Teacher Office Hours	by appointment
Online Course Link	https://ncueeclass.ncu.edu.tw/course/32506

2 Weekly Schedule

Week	Date	Topic
1	2026/02/26	Environment setup, variables, types, operators
2	2026/03/05	if/else, for/while loops, functions, scope, debugging
3	2026/03/12	Lists, dicts, tuples, sets; CSV & text file I/O
4	2026/03/19	PsychoPy install, Window, visual stimuli, clocks, input
5	2026/03/26	PsychoPy Experiment Design
6	2026/04/02	PsychoPy Builder, Stroop experiment, Pavlovia deployment
7	2026/04/09	NumPy arrays, vectorized operations, Matplotlib plots & subplots
8	2026/04/16	<i>Midterm: Deploy cognitive experiment to Pavlovia</i>
9	2026/04/23	Vibe/agent coding concepts; Claude Code CLI setup; prompting
10	2026/04/30	Multi-step task delegation, git commits, GitHub Actions
11	2026/05/07	Streamlit app; layout, widgets, charts; deploy to Streamlit Cloud
12	2026/05/12	Taiwan open data portal, API calls, data cleaning
13	2026/05/14	Interactive charts, narrative data storytelling
14	2026/05/21	Anthropic SDK; add AI features; responsible AI
15	2026/05/26	Final project workshop; peer code review; polish UI/UX
16	2026/05/28	<i>Final Presentation: Present online app</i>

3 Course Resources

- **GitHub Repository:**
<https://github.com/audachang/ai-programming-psychology>
- **Google Classroom:**
<https://classroom.google.com/c/ODQwMzEwNDQ3NjI5?cjc=hfwtvn66>

4 Grading Policy

Component	Description
Assignments (40%)	Four practical coding assignments: 1. PsychoPy Experiment Implementation 2. Statistical Analysis & Visualization 3. Machine Learning Model Training (Regression/Classification) 4. Advanced AI Pipeline (GPU/Deep Learning/LLM) All work must be synced to individual GitHub repositories.
Midterm Project (20%)	Experimental Demo: Demonstration of a custom-built psychological task using Python, focusing on creativity and technical precision.
Final Capstone (30%)	End-to-End AI Project: A complete pipeline from data source to AI prediction. Presented in a professional Poster Session format.
Participation (10%)	Code review contributions and active participation in class discussions.

5 Weekly Content Details

Module 1: Experimental Programming & Data Science

Focus: *From Hypothesis to Data* □ Mastering the Python ecosystem to build robust behavioral experiments, ensure reproducibility, and visualize scientific results.

Week 1: Orientation, Python Environment Setup & Basic Libraries

- **Date:** 2026/02/26

- **Topic: Setting the Stage for Scientific Computing**
- **Course Content:**
 - **The Stack:** VS Code, Anaconda/Miniconda, and managing Virtual Environments (`conda/venv`).
 - **Python Refresher:** Mutable vs. Immutable types, List Comprehensions, and Modular design.
 - **Scientific Libs:** Introduction to **NumPy** for high-performance array manipulation (essential for coordinate systems, timing, and randomization).
- **Teaching Plan:**
 - **Lecture:** The "Reproducibility Crisis" and how code provides the solution.
 - **Lab Activity (Install-Fest):** Ensure every student has a working local environment.
 - **Homework:** "Hello Data" □ Write a script that generates a synthetic dataset of reaction times using NumPy.

Week 2: PsychoPy Coder & Stimulus Presentation

- **Date:** 2026/03/05
- **Topic: Drawing to the Screen (Beyond the GUI)**
- **Course Content:**
 - **The Window:** `visual.Window`, units (`norm`, `pix`, `deg`), and frame buffering (`win.flip()`).
 - **Stimuli:** Programmatically creating `TextStim`, `ImageStim`, and `GratingStim` (Gabor patches).
 - **Timing:** Understanding refresh rates (Hz), frame-based timing, and precision.
- **Teaching Plan:**
 - **Live Coding:** Build a script from scratch that displays a fixation cross followed by a drifting Gabor patch.
 - **Lab:** Modify the script to change the orientation of the grating dynamically based on a frame counter.

Week 3: Interaction & The Event Loop

- **Date:** 2026/03/12
- **Topic: The Game Loop of Research**
- **Course Content:**

- **Input:** Polling (`defaultKeyboard.getKeys()`) vs. Blocking (`event.waitKeys()`).
- **Architecture:** Implementing the Trial → Block → Experiment hierarchy.
- **Data Logging:** Using `pandas` or `PsychoPy`’s `ExperimentHandler` to save trial-by-trial data to CSV.
- **Teaching Plan:**
 - **Lab Activity (The Stroop Task):** Build a complete interactive task where participants identify font colors of conflicting words.
 - **Crucial Step:** Students must run the task on themselves to generate real data for Week 5.

Week 4: PsychoPy Builder, Online Paradigms & Adaptive Design

- **Date:** 2026/03/19
- **Topic: Rapid Prototyping & Web Deployment**
- **Course Content:**
 - **Builder GUI:** Routines, Loops, and injecting Custom Code components.
 - **Pavlovia:** Converting Python experiments to PsychoJS for online data collection.
 - **Adaptive Design:** Introduction to **Staircase procedures** (e.g., Up/Down method) for finding sensory thresholds.
- **Teaching Plan:**
 - **Demo:** Recreate the Week 3 manual code using the Builder GUI in 15 minutes.
 - **Workshop:** Push a simple experiment to Pavlovia and collect data from a peer.

Week 5: PsychoPy Experiment Design

- **Date:** 2026/03/26
- **Topic: From Raw Logs to Insights**
- **Course Content:**
 - **Data Cleaning:** Using **Pandas** to filter outliers, group data, and handle missing values.
 - **Visualization:** **Seaborn** and **Matplotlib** for Violin plots, Scatter plots, and Error bars.
 - **Stats:** T-tests and correlations using `scipy.stats`.

- **Teaching Plan:**
 - **Data Dive:** Analyze the **Stroop Task data** generated in Week 3.
 - **Deliverable:** A script that produces a publication-quality figure showing the "Stroop Effect" (RT difference).

Week 6: PsychoPy Builder, Stroop Experiment & Pavlovia Deployment

- **Date:** 2026/04/02
- **Topic:** Rapid Prototyping with Builder & Online Deployment
- **Course Content:**
 - **Builder GUI:** Routines, Loops, Excel condition files, and injecting Custom Code components.
 - **Stroop Experiment:** Building a complete Stroop task in Builder – stimulus file design, keyboard responses, automatic scoring (`correct_key`), and pivot-table analysis.
 - **Code Components:** Independently randomizing words and colors so any word can appear in any ink color.
 - **Pavlovia:** Converting the experiment to PsychoJS for online data collection.
- **Teaching Plan:**
 - **Hands-on Workshop:** Build the Stroop experiment step-by-step in Builder and deploy to Pavlovia.
 - **Data Analysis:** Collect pilot data and summarize results using pivot tables (RT $\times$ congruency).

Week 7: NumPy, Data Manipulation & Visualization

- **Date:** 2026/04/09
- **Topic:** From Raw Data to Publication-Quality Figures
- **Course Content:**
 - **NumPy:** Array creation, indexing, broadcasting, vectorized operations, and boolean masking.
 - **Data Processing:** Reaction-time normalization (z-scores), outlier filtering, and batch processing with NumPy.
 - **Matplotlib:** Line plots, scatter plots, bar charts with error bars, histograms, multi-panel figures (subplots), and annotations.
- **Teaching Plan:**

- **Data Dive:** Analyze **Stroop task data** from Week 6 □ filter outliers with NumPy, compute condition means, and produce publication-quality figures showing the Stroop effect.

Week 8: Midterm Project Presentation

- **Date:** 2026/04/16
- **Activity:** Seminar & Demo
- **Requirements:**
 1. A fully functional Python/PsychoPy experiment.
 2. Automatic data logging.
 3. A visualization of pilot data.
 4. A clean GitHub repository link.

Module 2: Vibe/Agentic Coding & Data Applications

Focus: *From describing intent to shipping applications □ using AI as a coding partner to build, deploy, and narrate data-driven web apps grounded in real-world open data.*

Week 9: Vibe/Agentic Coding Fundamentals

- **Date:** 2026/04/23
- **Topic:** From Writing Code to Directing AI
- **Course Content:**
 - **The Spectrum:** Copilot autocomplete → Chat collaboration → Agentic execution.
 - **Vibe Coding Philosophy:** Describe intent, let AI scaffold, human verifies; comparing ChatGPT/Claude web, Cursor IDE, and Claude Code CLI.
 - **Setup:** Installing Node.js, Claude Code, API key management, first dialogue.
- **Teaching Plan:**
 - **Live Demo:** Have Claude Code read and improve a pandas notebook by prompt only.
 - **Lab:** Rewrite Week 7's Stroop analysis script using Claude Code; compare original vs. AI-assisted versions.
 - **Homework:** One-page reflection on where AI outperformed and where human judgment was essential.

Week 10: Multi-Step Task Delegation, Git & GitHub Actions

- **Date:** 2026/04/30
- **Topic:** Version Control in the Agentic Era
- **Course Content:**
 - **Prompt Design:** Plan–execute–verify pattern; `CLAUDE.md` and project context files.
 - **Git Essentials:** Repo, commit, branch, push, pull via GitHub Desktop (GUI) and Claude Code (CLI).
 - **GitHub Actions:** Automated checks on push (linting, Streamlit build verification).
- **Teaching Plan:**
 - **Demo:** Delegate an end-to-end pipeline (read CSV → clean → plot → commit → push) to Claude Code.
 - **Lab:** Create personal project repo on GitHub; configure a minimal GitHub Action.
 - **Homework:** Use AI to draft `README.md` and `CLAUDE.md` defining project goals and coding norms.

Week 11: Streamlit Fundamentals & Cloud Deployment

- **Date:** 2026/05/07
- **Topic:** From Script to Web App in Pure Python
- **Course Content:**
 - **Core API:** `st.write`, `st.dataframe`, widgets (`slider`, `selectbox`, `file_uploader`), layout (`columns`, `tabs`).
 - **State & Caching:** `st.session_state` and `@st.cache_data`.
 - **Deployment:** Streamlit Community Cloud with GitHub integration.
- **Teaching Plan:**
 - **Live Coding:** Vibe-code a Stroop analysis script into an interactive Streamlit app in 30 minutes.
 - **Lab:** Convert a personal pandas/matplotlib analysis into a deployed Streamlit app.
 - **Homework:** Submit deployed app URL.

Week 12: Taiwan Open Data & Real-World Cleaning

- **Date:** 2026/05/12
- **Topic:** APIs, Messy Data, and How AI Accelerates Cleaning
- **Course Content:**
 - **Data Sources:** data.gov.tw, TDX transport API, MOHW statistics, weather/air-quality APIs.
 - **HTTP/REST:** GET/POST, query parameters, JSON responses; `requests` with `pandas.read_json/read_csv`.
 - **Cleaning:** Chinese encoding, date parsing, missing values, deduplication, dtype coercion.
- **Teaching Plan:**
 - **Demo:** Claude Code analyzes a “dirty” air-quality dataset and proposes a cleaning pipeline with justifications.
 - **Lab:** Each group selects a behaviorally/socially relevant dataset; completes Download → Clean → EDA notebook.
 - **Homework:** Integrate the dataset into the Week 11 Streamlit app.

Week 13: Interactive Visualization & Data Storytelling

- **Date:** 2026/05/14
- **Topic:** Charts That Tell a Story
- **Course Content:**
 - **Storytelling Structure:** Hook → Context → Conflict → Insight → Conclusion.
 - **Tools:** `plotly.express`, Altair, `pydeck` for geospatial data.
 - **Design Principles:** Chart selection hierarchy, colorblind-safe palettes, misleading-axis anti-patterns.
- **Teaching Plan:**
 - **Comparison Demo:** Same data rendered as academic-style static figure vs. public-facing interactive chart.
 - **Lab:** Add ≥ 3 interactive elements (filters, tooltips, maps) and a narrative flow to the Streamlit app.
 - **Homework:** 3-minute demo video, practicing storytelling delivery.

Week 14: Anthropic SDK & Responsible AI Integration

- **Date:** 2026/05/21

- **Topic: Adding AI Features to Your App**
- **Course Content:**
 - **Anthropic API:** messages endpoint, system prompts, roles, streaming.
 - **Integration Patterns:** data summarization, Q&A over data, natural-language-to-pandas, classification/sentiment.
 - **Responsible AI:** hallucinations, privacy, de-identification, cost control, transparency (labeling AI output); alignment with EU AI Act and Taiwan’s AI Basic Act draft.
- **Teaching Plan:**
 - **Demo:** Add an “Ask this data” dialog to the Streamlit app □ natural language in, answer or chart out.
 - **Lab:** Each group integrates one AI feature (summary / chat / auto-report) with `.env`/Streamlit secrets for key management.
 - **Homework:** Add responsible-AI disclosure and API-cost estimate notes to the app.

Week 15: Final Project Workshop & Peer Code Review

- **Date:** 2026/05/26
- **Topic: From Working to Polished**
- **Course Content:**
 - **UI/UX Basics:** Hick’s law, Fitts’s law, consistency, visual hierarchy.
 - **Code Review Culture:** Structure of constructive feedback; reviewing repo layout, README clarity, app usability, AI feature design.
 - **Presentation Prep:** Narrative arc for a 10-minute demo.
- **Teaching Plan:**
 - **Short Lecture:** 30-minute overview of code review and UX principles.
 - **Workshop:** 150 minutes of open development; instructor circulates as technical consultant.
 - **Peer Review:** Paired 30-minute code review sessions; written feedback delivered before final presentation.

Week 16: Final Presentation

- **Date:** 2026/05/28
- **Activity: Symposium & Live Demo**

- **Deliverables:**

1. **Live Streamlit App:** Deployed online, integrating Taiwan open data and at least one AI feature.
2. **GitHub Repository:** Public repo with clear README, CLAUDE.md, and reproducible setup.
3. **10-Minute Presentation:** Problem framing, data, design decisions, AI usage, responsible-AI considerations, and live demo.